EXACT EXCLUSION OF 43,046,721 CROSS-BLOCK VOLTAGE GRAPHS FOR THE ERDŐS–GYÁRFÁS 2^k-CYCLE CONJECTURE

ARGUS OPEN RESEARCH PROJECT

Abstract. The Erdős–Gyárfás conjecture asserts that every graph of minimum degree at least 3 contains a simple cycle of length a power of 2. We prove that in the cross-block spoke voltage family over $\mathbb{Z}_{32}$, every one of the $3^{16} = 43{,}046{,}721$ voltage assignments contains a simple cycle of length 8, 16, or 32, covered by nine explicit cycle witnesses without symmetry reduction.

1. Main Family Exclusion

Theorem 1.1. *Across the* $3^{16} = 43{,}046{,}721$ *cubic voltage lifts in the cross-block spoke family over* $\mathbb{Z}_{32}$*, nine explicit closed voltage walk templates lift to simple cycles of length* 8*,* 16*, or* 32 *that partition the entire assignment space, proving that no graph in this family is a counterexample to the Erdős–Gyárfás conjecture.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1417 in the domain of graph theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped family exclusion* deciding the scope: all 3^{}16 = 43,046,721 cross-block spoke voltage assignments over Z_32 contain a C_8, C_16, or C_32 cycle.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- the unrestricted Erdos-Gyarfas conjecture.

References

[1] Primary problem list and source literature for Result 1417 (graph theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

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